

AdamW versus Muon: DistilBERT on SST-2

1. Which optimizer performed better, and by what criterion? Criterion: validation accuracy and loss after 3 matched epochs on SST-2 (67,349 train / 872 validation, batch size 32, seed 42, constant LR). AdamW (lr 2e-5 everywhere) reached final/best accuracy 90.83% (loss 0.26245, epoch 3). Muon (lr 0.003 on the 37 hidden weight matrices, AdamW lr 2e-5 fallback elsewhere) reached final accuracy 84.40% (loss 0.39393, epoch 3), best at epoch 1 (87.04%, loss 0.36956). By this criterion AdamW performed better here. Muon's matrix LR was fixed by a predeclared stability check over {0.001, 0.003, 0.01} on a disjoint 512/128 train/holdout split, selecting 0.003 as the smallest stable rate rather than the best-scoring one; 0.01 actually reached lower holdout loss in that check. So Muon's rate was chosen for stability, not tuned for accuracy, while AdamW's rate came from common practice for this model — the two optimizers were not given comparably tuned learning rates (full selection log in `SUPPLEMENTARY\_ANALYSIS.md`).

2. Which solution appears flatter? We perturbed each best checkpoint on 256 shared validation examples along 8 paired random directions per tensor, scaled to epsilon times each tensor's own norm (epsilon in {0.001,...,0.1}), and measured the mean loss increase — a local-curvature proxy: a flatter minimum tolerates larger perturbations with less loss increase. At epsilon=0.1, AdamW's mean increase was 0.00741 versus 0.00435 for Muon, suggesting Muon's checkpoint is locally flatter. But the paired Muon-minus-AdamW 95% directional interval is [-0.00976, 0.00365] — it includes zero, so the difference is not distinguishable from noise at this sample size. The two checkpoints are also from different epochs (AdamW epoch 3, Muon epoch 1) with different baseline losses (0.2253 vs 0.3199), so this is not a same-training-stage comparison.

3. How reliable are these conclusions? Limited. Each optimizer was run with one seed, so there is no estimate of run-to-run variance. Real per-step gradient/update norms were never recorded during the original 3-epoch runs; a separate, matched 32-step run (same init, batch order, RNG state) is used instead to characterize update dynamics, and must not be read as reconstructing the actual training trajectory. In that 32-step run, Muon's matrix-group update norm rose over the first ~5 steps then plateaued near 0.49, while AdamW's started near 0.13 and decayed toward 0.05; matrix-group gradient norms fluctuated similarly for both (no sustained divergence), so the gap is a property of each optimizer's update rule (Muon's orthogonalized step versus AdamW's per-parameter adaptive scaling), not of gradient magnitude. Both optimizers use $\text{weight_decay}=0.01$, applied as a decoupled multiplicative step ($p \leftarrow p \cdot (1 - \text{lr} \cdot \text{weight_decay})$); with $\text{lr}=0.003$ for Muon's matrix group versus $\text{lr}=2\text{e-}5$ for AdamW, the realized per-step decay is about 150x larger for Muon's matrix parameters, confounding the update-rule and effective-regularization differences. Muon's full run also took 3621s versus AdamW's 1603s (~2.26x) on this Apple M4/MPS machine with this reference (unfused, per-parameter Newton-Schulz) implementation — a cost specific to this hardware/implementation pairing, not a general Muon-vs-AdamW throughput claim. See `SUPPLEMENTARY\_ANALYSIS.md` for the full 32-step trend and timing discussion.

4. What would you change in a larger study? Run multiple seeds per optimizer to quantify variance in accuracy, loss, and sharpness. Record real per-step gradient/update norms during the full runs instead of a separate short proxy run. Evaluate sharpness at matched checkpoint epochs (or matched step counts) so the comparison is not confounded by different training stages. Tune AdamW's learning rate with the same predeclared, stability-first procedure used for Muon, and consider matching effective weight decay ($\text{lr} \times \text{weight_decay}$) rather than the nominal coefficient. Time both optimizers on the same hardware with a fused/optimized Muon implementation before drawing any throughput conclusion.